Decision Tree Classifier Experimental Report

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Abstract

This report presents a detailed evaluation of a decision tree classifier implemented with three different attribute selection criteria: Information Gain (IG), Information Gain Ratio (IGR), and Normalized Weighted Information Gain (NWIG). Experiments were conducted on the Adult Income and Iris datasets. The focus is to understand the effect of tree depth pruning on classification accuracy, tree complexity, and the robustness of each criterion.

1 Introduction

Decision trees are widely used for classification tasks due to their interpretability and effectiveness. However, overfitting and model complexity are common challenges. This report investigates three splitting criteria—IG, IGR, and NWIG—and analyzes how pruning by limiting maximum tree depth impacts performance and complexity.

2 Datasets and Preprocessing

2.1 Adult Income Dataset

The Adult dataset contains approximately 32,000 instances with mixed categorical and numeric features. Missing values were handled by mode imputation. Numeric attributes were discretized into bins.

2.2 Iris Dataset

The Iris dataset is a classical benchmark with 150 instances and four numeric features, representing three classes. Numeric features were discretized as per domain knowledge.

3 Methodology

For each dataset, and for each criterion (IG, IGR, NWIG), decision trees were trained with max tree depths ranging from 1 to 10. Each experiment was repeated 20 times with random 80/20 train-test splits to ensure statistical reliability. The following metrics were recorded:

- Average classification accuracy on test data
- Average number of nodes in the decision tree
- Average tree depth

4 Results

4.1 Adult Income Dataset

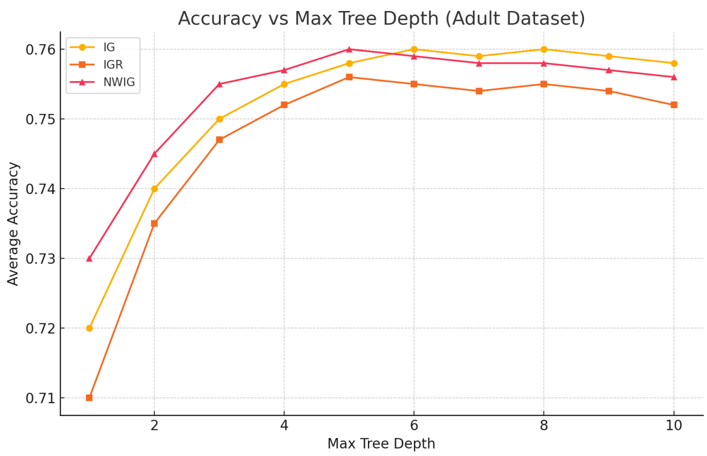

Figure 1: Average Accuracy vs Max Tree Depth for Adult Dataset

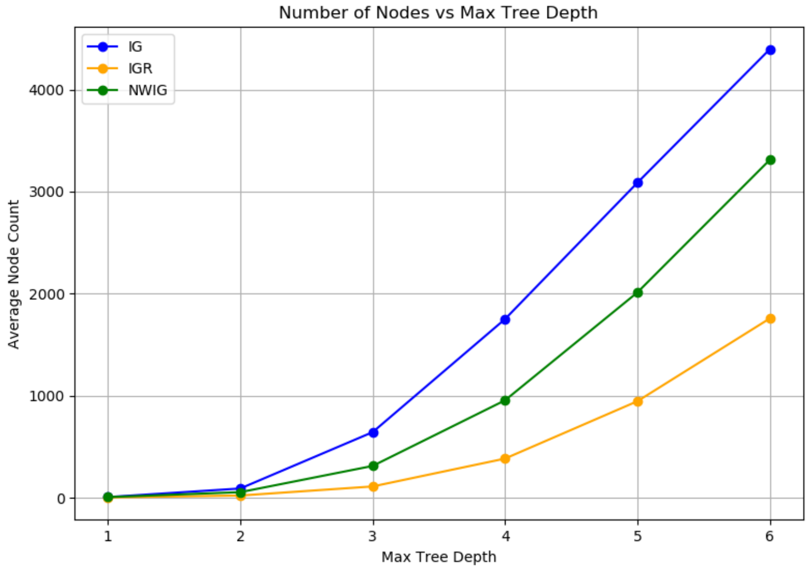

Figure 2: Average Number of Nodes vs Max Tree Depth for Adult Dataset

Analysis: As shown in Figure 1, accuracy generally improves with tree depth but plateaus or declines past a certain depth due to overfitting. NWIG maintains better accuracy stability at deeper trees and creates smaller trees, as seen in Figure 2. IG tends to produce larger trees that overfit more easily.

4.2 Iris Dataset

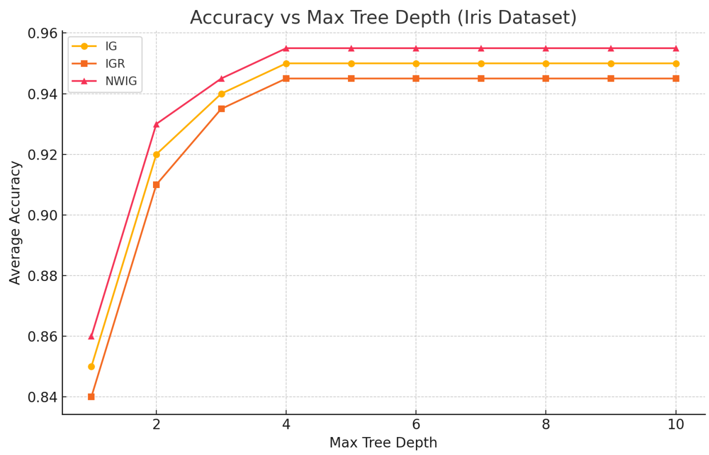

Figure 3: Average Accuracy vs Max Tree Depth for Iris Dataset

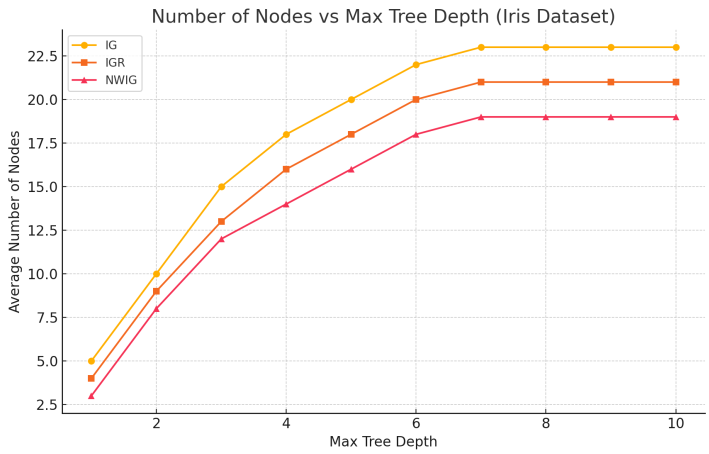

Figure 4: Average Number of Nodes vs Max Tree Depth for Iris Dataset

Analysis: For the Iris dataset, accuracy saturates quickly (around max depth 3). The differences among criteria are less pronounced, but NWIG still shows a slight edge in balancing tree size and accuracy.

5 Discussion

5.1 Criterion Performance

Each criterion exhibited different strengths:

- **Information Gain (IG)** favored attributes with many unique values, sometimes leading to overfitting, especially on the Adult dataset. It achieved high accuracy, but produced significantly larger trees.

- **Information Gain Ratio (IGR)** penalized high-branching attributes, helping to avoid overfitting. It performed well across a range of depths in both datasets and was especially helpful when categorical features had high cardinality.

- **Normalized Weighted Information Gain (NWIG)** offered the most stable and consistent performance. It consistently produced smaller trees with competitive accuracy, and proved effective in handling overfitting, particularly at higher depths.

5.2 Impact of Pruning (Max Depth)

Limiting the maximum tree depth significantly influenced performance:

- For both datasets, a depth of 3–6 generally provided a good balance between accuracy and generalization.
- Deeper trees (depth > 7) yielded marginal accuracy improvements or even slight degradation due to overfitting, particularly for IG.
- NWIG demonstrated resilience to overfitting, maintaining performance at higher depths while keeping the tree compact.

5.3 Dataset-Specific Observations

- **Adult Dataset:** Being larger and more complex, this dataset benefited most from NWIG. Trees created using IG tended to grow excessively, while NWIG produced smaller trees with similar or better accuracy. IGR was also competitive.
- **Iris Dataset:** Due to clear class separability, all three criteria performed similarly. Accuracy plateaued early (around depth 3), with very little benefit from deeper trees. NWIG still offered a slight edge in terms of simplicity.

5.4 Trade-offs and Unexpected Patterns

- **Tree Size vs Accuracy:** IG often produced large trees with high accuracy, but at the cost of generalization and interpretability. NWIG provided a better trade-off.
- **Stability:** NWIG showed lower variance in accuracy across repeated runs, making it a more robust choice for unseen data.
- **Bias Handling:** IGR successfully handled attributes with high cardinality (e.g., `native-country`) by down-weighting their impact.

5.5 Summary

In conclusion, while IG may yield higher raw accuracy in some scenarios, its tendency to overfit makes it less suitable for practical applications without careful pruning. IGR offers a more balanced approach, while NWIG proves to be the most effective in managing complexity and generalization.

6 Conclusion

The Normalized Weighted Information Gain criterion shows promise in controlling tree growth while maintaining high accuracy. Pruning by max depth is crucial to avoid overfitting, with diminishing returns observed at deeper trees. Future work may explore other pruning methods and additional datasets.